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FROM ZERO TO AGENTS: DECONSTRUCTING THE AI REVOLUTION

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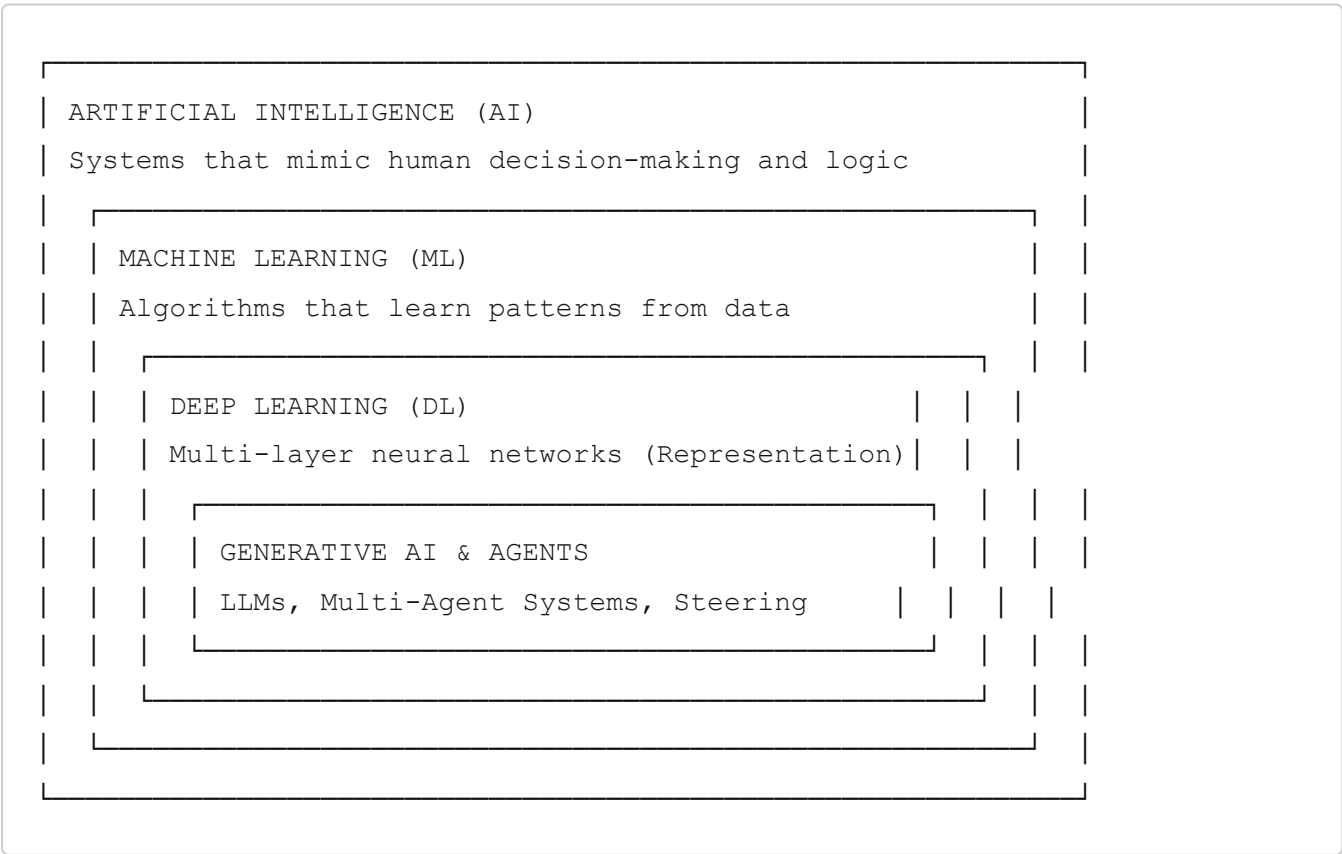
July 23, 2026 [junaidte14](https://blog.codoplex.com/author/junaidte14gmail-com/) (<https://blog.codoplex.com/author/junaidte14gmail-com/>) 2–3 minutes

Welcome & Course Blueprint: How We Will Build Modern AI from Scratch (Ep.001)

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At its core, **Artificial Intelligence (AI)** is the discipline of creating algorithms that allow computers to perform tasks that previously required human cognitive abilities—such as reasoning, recognizing patterns, making decisions, and processing natural language.

While “AI” sounds like a single entity, it is actually a hierarchy of nested concepts:



2. The Core Branches: Vision vs. Text (Language)

Modern Deep Learning generally splits into two major sensory domains, each with its own underlying mathematical structures:

Domain	Primary Modality	Core Math Primitive	Architectural Benchmark	Primary Use Cases
Computer Vision (CV)	Spatial grid arrays (Pixels)	2D/3D Convolutions ($\ast$), Spatial Pooling	CNNs, Vision Transformers (ViT), Diffusion	Image recognition, medical imaging, video synthesis
Natural Language (NLP)	Discrete, sequential symbolic tokens	Vector Embeddings, Sequence Gating, Self-Attention (Q, K, V)	RNNs, GRUs, Transformers, LLMs	Machine translation, code generation, reasoning, Agentic AI

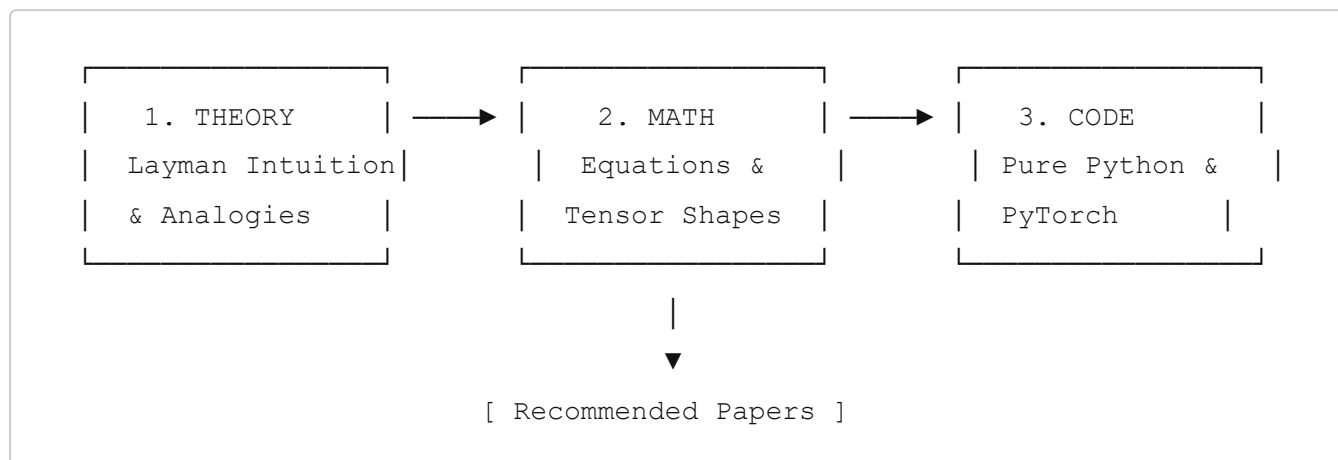
Why Our Focus is Text & Sequential Intelligence

While image generation is visual and expressive, **language is the native substrate of reasoning, logic, and planning.**

By focusing on text and sequence modeling, we gain direct access to the exact primitives that make modern **Large Language Models (LLMs)** and **Agentic AI** possible. Understanding how a model handles sequence dependencies gives you the keys to building reasoning engines, code assistants, and autonomous agents.

3. Our 3-Pillar Learning Framework

Every single lesson in this course is designed around three distinct, non-negotiable pillars:



- 1. Theoretical Intuition (Layman Perspective):** We start with high-level conceptual analogies and the historical problem the creators were trying to solve.
- 2. Mathematical Formulation:** We strip away black-box abstractions and break down the exact linear algebra, calculus, and probability equations powering the system.
- 3. From-Scratch Python/PyTorch Code:** We implement the mechanism step-by-step—first using raw Python lists and loops to see the gears turning, then converting it to production-grade PyTorch tensors.
- 4. Curated Paper Reading:** Each post concludes with the essential academic papers that defined that specific breakthrough, giving you a researcher-level bibliography.

4. The Curriculum Roadmap: From Vectors to Autonomous Agents

We will proceed step-by-step without rushing, taking one core module at a time:

- **Module 1: Foundations of Representing Language**
 - *Post 1.1:* Language as Geometry (Vectors, Dot Products & Matrix Multiplication)
 - *Post 1.2:* Non-Linearity & Classification Heads (Sigmoid, Tanh, Softmax)
- **Module 2: The Evolution of Sequence Memory**
 - *Post 2.1:* Simple Recurrent Neural Networks (RNNs) & Vanishing Gradients
 - *Post 2.2:* Gated Recurrent Units (GRUs) & Memory Highways

- **Module 3: The Transformer Paradigm**

- *Post 3.1:* The Bottleneck of Sequential Processing & Scaled Dot-Product Attention
- *Post 3.2:* Multi-Head Attention (MHA) & Positional Encodings (Absolute vs. RoPE)
- *Post 3.3:* Encoders, Decoders & The Complete Transformer Architecture

- **Module 4: Modern LLM Dynamics & Interpretability**

- *Post 4.1:* Normalization (LayerNorm vs. RMSNorm) & Residual Stream Mechanics
- *Post 4.2:* Mechanistic Interpretability: Probing, Neuron Activation Maps & Monosemantic Features

- **Module 5: Agentic Systems & Human Steering**

- *Post 5.1:* Dense Retrieval, Graph-RAG & Hybrid Search Architectures
- *Post 5.2:* Human-in-the-Loop Steering, Active Learning & Multi-Agent Orchestration

5. Recommended Foundational Reading

Before diving into Post 1, explore this classic survey to contextualize where language technology started:

- **Jurafsky & Martin:** *Speech and Language Processing (3rd ed. draft)*.
(<https://web.stanford.edu/~jurafsky/slp3/>) — Chapter 6: Vector Semantics and Embeddings.

Ready to Begin?

Now that our blueprint is established, we can jump right into **Module 1, Post 1: Language as Geometry — Vectors, Dot Products & Matrix Transformations!**

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